

1.Explain the importance and need for network security in modern systems.

Network security is the process of protecting computer networks, devices, and data from unauthorized access, misuse, attacks from the hacker.

Importance of Network Security

1. **Protects Data and Information:** it protect the data such as passwords bank details personal files etc
2. **Prevents Unauthorized Access:** It ensures that only authorized users can access network resources and services
3. **Maintains Confidentiality:** Important information remains private and is not disclosed to outsiders.
4. **Protects Against Cyber Attacks:** It defends networks from viruses, malware
5. **Builds User Trust:**

Need for Network Security in Modern Systems

1. **Increased Internet Usage**
2. **Growth of Online Services**
4. **Protection of Financial Transactions**

2.What are different security principles and security services?

Security Principles

Security principles are the basic rules used to protect computer networks and information systems.

1. Confidentiality

Information should be accessible only to authorized users.

2. Integrity

Data should not be modified or changed by unauthorized persons.

3. Availability

Network resources and services should be available to authorized users whenever needed.

4. Authentication

Verifies the identity of users or devices.

5. Authorization

Determines what resources a user is allowed to access.

Services:1.Authentication service

2.access control service

3.data integrity service 4.availability service.

3. differentiate between passive attack and active attack.

Passive Attack

1. The attacker only monitors the communication.
2. No changes are made to data.
3. Difficult to detect.
4. Affects confidentiality of data.
5. Does not harm the system directly.
6. Prevention is more important than detection.
7. Examples: Eavesdropping, Traffic Analysis

Active Attack

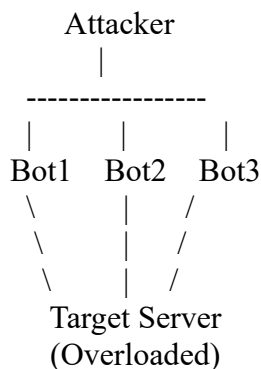
- The attacker modifies or damages the communication.
- Data or system resources are altered.
- Easier to detect.
- Affects integrity and availability of data.
- Can cause damage to the system or network.
- Detection and recovery are important.
- Examples: DoS attack, Masquerade attack, Message modification

4. Explain DDoS attack with a diagram.

DDoS stands for distributed denial of service attack, A DDoS attack is a type of cyber attack in which multiple bots computers send a huge amount of traffic or requests to a target server or network, making it unavailable to legitimate users.

It is a major threat to network availability and security

DIAGRAM:



Effects of DDoS Attack

- Server downtime
- Slow network performance
- Loss of business services
- Financial loss

Prevention Methods

- ☐ Firewall protection
- ☐ Traffic filtering , Load balancing , Anti-DDoS services

6. Explain Block cipher modes in detail.

Block cipher modes are techniques used to encrypt data in blocks using a block cipher algorithm such as DES or AES.

Types of Block Cipher Modes:

1. Electronic Code Book (ECB) Mode:

ECB is a block cipher mode in which each plaintext block is encrypted independently using the same secret key.

Advantages

- Simple and fast.
- Easy implementation.

Disadvantages

- Identical plaintext blocks produce identical ciphertext blocks.
- Less secure.

2. Cipher Block Chaining (CBC) Mode

CBC is a block cipher mode in which each plaintext block is XORed with the previous ciphertext block before encryption.

Advantages

- More secure than ECB.
- Hides data patterns.

Disadvantages

- Encryption is slower.
- Error propagation occurs.

2. Cipher Feedback (CFB) Mode

CFB is a block cipher mode that converts a block cipher into a stream cipher by feeding previous ciphertext blocks back into the encryption process.

Advantages

- Suitable for real-time communication.
- No need for padding.

Disadvantages

- Slower process.
- Error affects current and next block.

3. Output Feedback (OFB) Mode

OFB is a block cipher mode in which encrypted output is repeatedly fed back to generate a keystream for encryption and decryption.

Advantages

- No error propagation.
- Suitable for noisy channels.

Disadvantages

- Synchronization required

7. Compare CFB and OFB modes.

CFB (Cipher Feedback Mode)

Converts block cipher into stream cipher using ciphertext feedback.

Previous ciphertext block is fed back into encryption.

Ciphertext depends on both key and plaintext.

Error in one block affects current and next block.

Suitable for real-time communication.

Encryption process is sequential.

OFB (Output Feedback Mode)

Converts block cipher into stream cipher using output feedback.

Previous output of encryption is fed back into encryption.

Keystream depends only on key and IV.

Error in one block affects only that block.

Suitable for noisy communication channels.

Keystream can be generated in advance.

8. What is substitution cipher? Explain the mono alphabetic cipher and poly alphabetic cipher.

A substitution cipher is an encryption technique, in which each letter or group of letters in the plaintext is replaced by another letter, symbol, or character to produce ciphertext.

Monoalphabetic Cipher:

A monoalphabetic cipher is a substitution cipher in which each plaintext letter is always replaced by the same ciphertext letter throughout the message.

Example: plaintext: hello ciphertext:itssg

Advantages

- Simple and easy to implement.
- Faster encryption process.

Disadvantages

- Less secure.
- Can be broken using frequency analysis.

Polyalphabetic Cipher:

A polyalphabetic cipher is a substitution cipher in which multiple substitution alphabets are used for encryption.

Advantages

- More secure than monoalphabetic cipher.
- Difficult to break using frequency analysis.

Disadvantages

- More complex.
- Requires key management.

8. Describe Man-in-the-Middle attack and its impact.

A Man-in-the-Middle attack is a cyber attack in which an attacker secretly stands between two communicating users and intercepts or changes their data without their knowledge.

Sender $\longleftrightarrow$ Attacker $\longleftrightarrow$ Receiver

Impact of MITM Attack

1. Data Theft

Sensitive information such as passwords, bank details, and personal data can be stolen.

2. Loss of Privacy

Private communication becomes exposed to attackers.

3. Financial Loss

Attackers may misuse banking or payment information.

4. Data Modification

Messages or files can be altered during transmission

10. Explain the CIA Triad (Confidentiality, Integrity, Availability).

The CIA Triad is a basic security model used in computer networks and information security. It consists of three important principles: Confidentiality, Integrity, and Availability.

1. Confidentiality: Confidentiality means protecting data from unauthorized access or disclosure.
2. Integrity: integrity means ensuring that data remains accurate, complete and unchanged
3. Availability: availability means ensuring that data and network resources are accessible to authorized users whenever needed

11. What is stream Cipher? Explain the encryption process using stream Cipher with suitable example.

A stream cipher is an encryption technique in which plaintext data is encrypted one bit or one byte at a time using a secret key and a keystream.

Encription process:

- ☐ Plaintext is taken one bit/byte at a time.
- ☐ A keystream is generated using a secret key.
- ☐ Plaintext bits are combined with keystream bits using XOR operation.
- ☐ The result produced is ciphertext.

Example: Plaintext (P) : 10101010

Keystream (K): 11001100

Encryption using xor

10101010

$\oplus$ 11001100

01100110

Advantages

- Fast encryption process
- Suitable for real-time communication